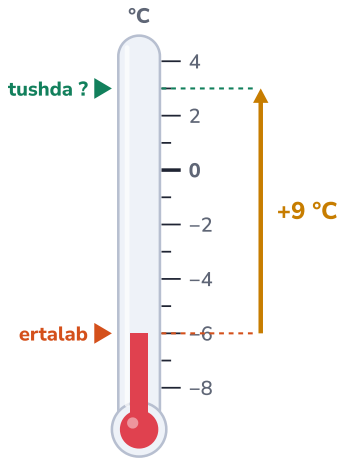
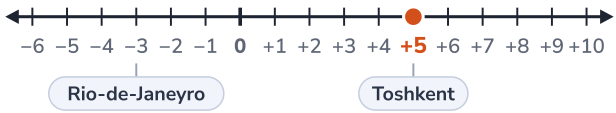
Butun sonlarni qo'shish va ayirish

1-QISM

Qiziqtiramiz

IKKI SAVOL





Vaqt mintaqalari

1-savol. Qishning sovuq kunida ertalab ko'cha termometri −6 °C ni ko'rsatdi. Tushgacha havo 9 °C ga isidi. Tushda termometr necha gradusni ko'rsatadi?

2-savol. Toshkent GMT+5, Rio-de-Janeyro (Braziliya) GMT−3 vaqt mintaqasida. Toshkentda soat 20:00 bo'lganda Rio-de-Janeyroda soat nechi?



−6 va 9 ni qo'shsam, 15 chiqadimi yoki 3? Vaqt farqi esa 5 − 3 = 2 soatmi?

ESLAB OLAYLIK

- Butun sonlar: ..., −3, −2, −1, 0, 1, 2, 3, ... Manfiy sonlar sonlar o'qida 0 dan chapda turadi.
- Kasr yoki o'nli kasr ko'rinishida yozilgan son ham butun bo'lishi mumkin: $\frac{10}{5} = 2$, −4,0 = −4. Lekin 2,5 va $\frac{1}{3}$ — butun emas.
- 7 va −7 — qarama-qarshi sonlar: ular 0 dan bir xil uzoqlikda, lekin har xil tomonda.
- Haroratda «0 dan past» — manfiy, pulda «chiqim» yoki «qarz» — manfiy son bilan yoziladi.

2-QISM

Sinab ko'ramiz

1-TOPSHIRIQ · TUGMALAR

Sariq tugma $+1$ ni, ko'k tugma -1 ni bildiradi. Rasmlarda qaysi son tasvirlangan?

a)



b)



c) Bitta sariq va bitta ko'k tugma birgalikda nechaga teng? Uchta sariq va uchta ko'k-chi?

2-TOPSHIRIQ · TUGMALARNI BIRLASHTIRING

Har bir yig'indini tugmalar bilan tasavvur qiling. Bir sariq va bir ko'k tugmani juft qilib olib tashlang. Nechta tugma qoldi va qaysi rangda?

a) $(-4) + (-2)$ b) $6 + (-2)$ c) $(-6) + 2$

3-TOPSHIRIQ · OLIB TASHLASH

Stolda 2 ta sariq tugma bor (bu 2 soni). a) Undan 5 ta sariq tugmani olib tashlash kerak ($2 - 5$). Tugma yetmayapti — nima qilish mumkin? Yordam: bir sariq va bir ko'k tugmani qo'shsak, stoldagi son o'zgarmaydi.
b) 2 ta sariq tugmadan 3 ta ko'k tugmani olib tashlash kerak ($2 - (-3)$). Nima qolishi mumkin?

4-TOPSHIRIQ · SOLISHTIRING

Har juftni hisoblang. Nimani sezdingiz?

a) $9 - 4$ va $9 + (-4)$ b) $2 - (-3)$ va $2 + 3$ c) $4 - 9$ va $9 - 4$

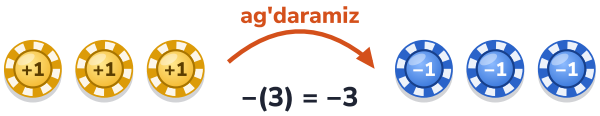
3-QISM

Tushunamiz

TA'RIF · ALGEBRAIK TUGMA

Musbat va manfiy sonlarni **algebraik tugmalardan** foydalanib ham ko'rsatish mumkin. Har bir algebraik tugmaning ikkita tomoni bor. Uning bir tomonida «1», ikkinchi tomonida «−1» yozilgan.

Songa **qarama-qarshi sonni** hosil qilish uchun algebraik tugmani teskarisiga ag'darib qo'yamiz. Bunda $-(1) = -1$ va $-(-1) = 1$ bo'ladi.



Musbat son $+1$ tugmalar bilan, manfiy son -1 tugmalar bilan tasvirlanadi: $1 + 1 + 1 + 1 + 1 = 5$; $(-1) + (-1) + (-1) + (-1) = -1 - 1 - 1 - 1 = -4$.

TA'RIF · NOL JUFTLIK

1 va −1 algebraik tugmalardan 0 hosil bo'ladi. Ular **nol juftlik** deb ataladi: $1 + (-1) = 1 - 1 = 0$.

Butun sonlarni qo'shish amali **kommutativdir**, ya'ni butun sonlarni qo'shishda ularning o'rnini almashtirgan bilan yig'indi o'zgarmaydi: $(-8) + 3 = 3 + (-8)$.

USUL · BUTUN SONLARNI ALGEBRAIK TUGMALAR YORDAMIDA QO'SHISH

1. Birinchi qo'shiluvchini tugmalar bilan tasvirlaymiz.
2. Ikkinchi qo'shiluvchining tugmalarini qo'shamiz.
3. Nol juftliklarni olib tashlaymiz. Qolgan tugmalar — yig'indi.

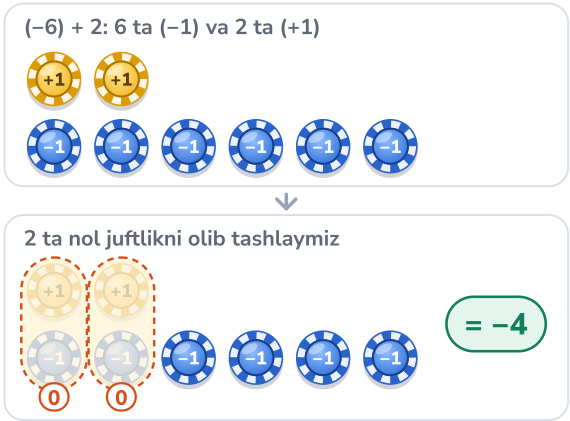
Qo'shiluvchilar manfiy birlardan iborat bo'lsa, yig'indining ishorasi ham manfiy bo'ladi. Musbat 1 lar soni manfiy 1 lar sonidan ko'p bo'lsa, yig'indi musbat; manfiy 1 lar ko'p bo'lsa — manfiy bo'ladi.

1-MISOL · ISHORALARI BIR XIL



$(-4) + (-3) = -4 - 3 = -7$

2-MISOL · ISHORALARI HAR XIL



$(-6) + 2 = -6 + 2 = -4$. Yoki o'rnini almashtirib: $2 + (-6) = 2 - 6 = -4$.

Katta sonlarda tugma chizib o'tirmaymiz, faqat ishorani aniqlab hisoblaymiz: $31 + (-18) = 31 - 18 = 13$ — musbatlar ko'p; $(-25) + 9 = -25 + 9 = -16$ — manfiylar ko'p.

TA'RIF · QARAMA-QARSHI SON

Sonning ishorasini o'zgartirish orqali shu songa qarama-qarshi songa ega bo'lamiz. Masalan, 6 ga qarama-qarshi son: −6. −6 ga qarama-qarshi son: $-(-6) = 6$.

USUL · BUTUN SONLARNI ALGEBRAIK TUGMALAR YORDAMIDA AYIRISH

1. Kamayuvchini tugmalar bilan tasvirlaymiz.
2. Ayiriluvchiga teng tugmalarni olib tashlaymiz.
3. Olib tashlashga kerakli tugmalar yetmasa — avval nol juftliklar qo'shamiz (son o'zgarmaydi). Istalgancha nol juftlik qo'shish mumkin, lekin yetmagan tugmalar sonicha qo'shish kifoya.

3-MISOL · $2 - 5$



$2 - 5 = 2 + (-5) = -3$

4-MISOL · $1 - (-4)$



$1 - (-4) = 1 + 4 = 5$

5-MISOL · $(-1) - 3$

1 ta (-1) tugma bor, $(+1)$ tugma yo'q. 3 ta nol juftlik qo'shamiz va 3 ta $(+1)$ ni olib tashlaymiz. 4 ta (-1) qoladi:

$(-1) - 3 = -1 - 3 = -4$

6-MISOL · $(-6) - (-2)$

6 ta (-1) tugmadan 2 tasini olib tashlaymiz — tugmalar yetadi, nol juftlik kerak emas. 4 ta (-1) qoladi:

$(-6) - (-2) = -6 + 2 = -4$

QOIDA · AYIRISH VA QARAMA-QARSHI SON

Biror sonni ayirishni sonning qarama-qarshisini qo'shish sifatida ifodalashimiz mumkin. Masalan, $2 - 7 = 2 + (-7)$ (2 ayiruv 7 — 2 qo'shuv manfiy 7).

Manfiy sonning qarama-qarshisi yoki manfiysi musbat son bo'ladi. Masalan, $-(-8) = 8$ (manfiy 8 ning manfiysi — musbat 8).

7-MISOL · MUSBAT SONNI AYIRISH

$8 - 15 = 8 + (-15) = -7$ — manfiy birlar ko'p.

$-12 - 9 = -12 + (-9) = -21$

8-MISOL · MANFIY SONNI AYIRISH

$-9 - (-14) = -9 + 14 = 5$

$0 - (-6) = 0 + 6 = 6$

GRINVICH VAQTI

Grinvich vaqti (GMT) — Angliyadagi Grinvich rasadxonasidan o'tuvchi bosh meridian bo'yicha hisoblanadigan vaqt. O'zbekiston undan 5 soat oldinda — GMT+5. Ikki joy orasidagi vaqt farqi — vaqt mintaqalarining ayirmasi.

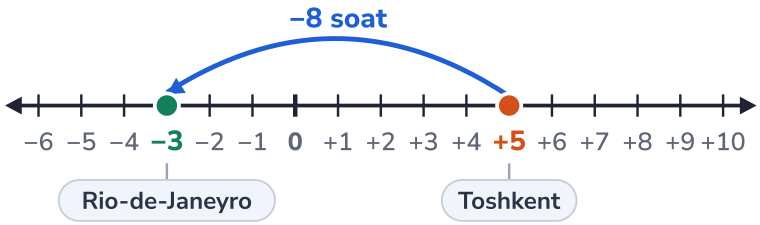
4-QISM

Qo'llaymiz

1-QISMDAGI SAVOLLAR

1-savol. $(-6) + 9 = -6 + 9 = 3$ °C (15 emas: manfiy va musbat birlar bir-birini yo'q qiladi).

2-savol. Mintaqalar ayirmasi: $5 - (-3) = 5 + 3 = 8$ soat (2 emas). Toshkent 8 soat oldinda: $20{:}00 - 8$ soat $= 12{:}00$.



1-MISOL · ISHORANI OLDINDAN AYTISH

Javob musbatmi yoki manfiymi? So'ng hisoblang: a) $(-13)+(-9)$ b) $27+(-8)$ c) $(-21)+6$ d) $0+(-7)$

a. Ikkalasi manfiy — yig'indi manfiy: $-13-9=-22$.

b. Musbat birlar ko'p (27 > 8) — musbat: $27-8=19$.

c. Manfiy birlar ko'p (21 > 6) — manfiy: $-21+6=-15$.

d. Faqat manfiy birlar — manfiy: $0-7=-7$.

2-MISOL · UCH HADLI IFODA

$-6-(-11)+(-9)$ ni hisoblang.

1. Manfiy sonni ayirish — qarama-qarshisini qo'shish: $-(-11)=+11$. Manfiy sonni qo'shish — ayirish: $+(-9)=-9$.
2. $-6+11-9$: chapdan o'ngga: $-6+11=5$, keyin $5-9=-4$.

3-MISOL · TUSHIRIB QOLDIRILGAN SON

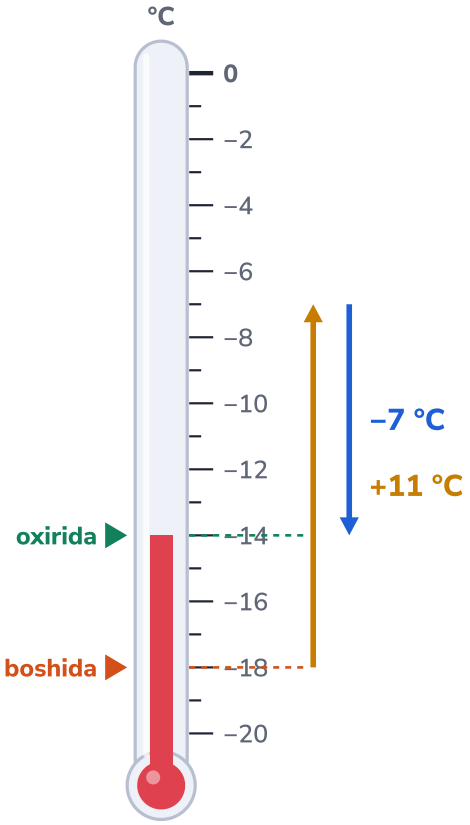
$-7-\square=4$. $\square$ ni toping.

1. Ayiriluvchi = kamayuvchi − ayirma: $\square=-7-4$.
2. $-7-4=-11$.
3. ✓ $-7-(-11)=-7+11=4$.

Agar $\square$ qo'shilgan bo'lsa ($-7+\square=4$), unda $\square=4-(-7)=11$ bo'lardi.

4-MISOL · MUZLATKICHDAGI HARORAT

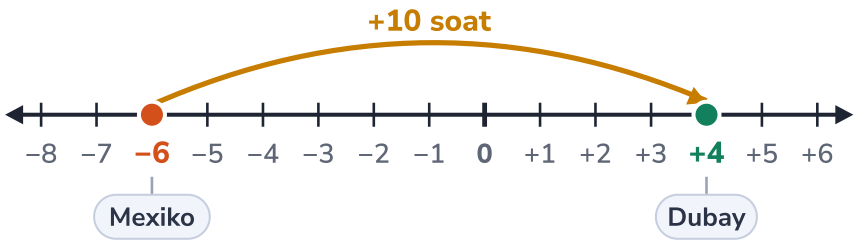
Muzlatish kamerasida harorat −18 °C edi. Tok o'chib qolib, harorat 11 °C ga ko'tarildi. Tok kelgach, kamera haroratni 7 °C ga pasaytirdi. Endi kamerada necha gradus?



1. Ko'tarilish — musbat, pasayish — manfiy son: $-18 + 11 + (-7)$.
2. $-18 + 11 = -7$, keyin $-7 - 7 = -14$ °C.

5-MISOL · PARVOZ VA VAQT MINTAQALARI

Dubay GMT+4, Mexiko GMT−6 vaqt mintaqasida. Dubaydan uchgan samolyot 16 soat yo'l yurib, Mexikoga mahalliy vaqt bilan 05:00 da qo'ndi. Samolyot Dubay vaqti bilan soat nechada uchgan edi?

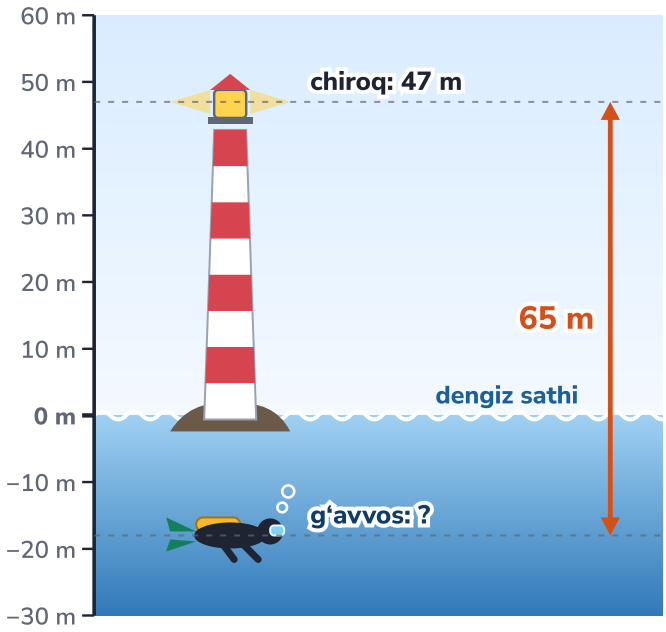


1. Vaqt farqi: $4 - (-6) = 4 + 6 = 10$ soat — Dubay 10 soat oldinda.

2. Qo'ngan payt Dubay vaqtida: $05{:}00 + 10$ soat $= 15{:}00$.
3. 16 soat orqaga qaytamiz: $15{:}00 - 15$ soat $= 00{:}00$, yana 1 soat — oldingi kun $23{:}00$.

6-MISOL · BALANDLIK VA CHUQURLIK

Qoyadagi mayoq chirog'i dengiz sathidan 47 m balandda. G'avvos chiroqdan roppa-rosa 65 m pastda suzmoqda. G'avvos dengiz sathidan necha metr chuqurlikda?



1. Balandlik — musbat son: chiroq $+47$. 65 m pastga tushish — 65 ni ayirish.
2. $47 - 65 = 47 + (-65) = -18$: g'avvos 18 m chuqurlikda.

✓ Ikki joy orasidagi tik masofa — yuqoridagi sondan pastdagisini ayiramiz: $47 - (-18) = 47 + 18 = 65$ m.

7-MISOL · ROSTMI YOKI YOLG'ON?

a) «Manfiy son + musbat son = manfiy son». b) «Ikki qarama-qarshi sonning yig'indisi 0 ga teng».

a. Yolg'on. **Qarshi misol** (tasdiqning yolg'on ekanini ko'rsatuvchi misol): $(-2) + 6 = 4$ — musbat.

b. Rost: masalan, $9 + (-9) = 0$ — hammasi nol juftliklarga ajraladi.



Ikki minus yonma-yon kelsa — $-(-a)$ — plyusga aylanadi. Lekin $-5-3$ da minuslar yonma-yon emas!

DIQQAT!

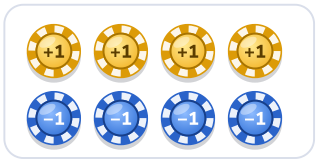
1. Manfiy sonni ayirish: ✗ $4-(-6)=-2$ ✓ $4-(-6)=4+6=10$
2. Ikki manfiy son yig'indisi manfiy: ✗ $(-8)+(-2)=10$ ✓ $(-8)+(-2)=-10$
3. Manfiydan musbatni ayirish: ✗ $-5-3=-2$ ✓ $-5-3=-8$
4. Kichikdan kattani ayirish: ✗ $3-7=4$ ✓ $3-7=-4$
5. Ayirishda o'rin almashmaydi: ✗ $4-9=9-4$ ✓ $4-9=-5$, lekin $9-4=5$

Mashqlar

1. Qaysilari butun son? $\frac{12}{4}$; −5; 3,6; 0; $-2\frac{1}{2}$; −9,0; $\frac{7}{3}$; 100
2. Tugmalar qaysi sonni tasvirlaydi?



a)



b)



c)

3. Qarama-qarshi sonni yozing: a) 14 b) −23 c) 0 d) $-(-9)$
4. Hisoblang.
   a) $9+6$ b) $(-7)+(-8)$ c) $(-12)+5$ d) $15+(-6)$
   e) $(-4)+11$ f) $0+(-19)$ g) $(-33)+33$ h) $26+(-40)$
5. Hisoblang.
   a) $4-11$ b) $(-9)-6$ c) $8-(-5)$ d) $(-14)-(-20)$
   e) $(-7)-(-7)$ f) $-15-17$ g) $(-22)-(-13)$ h) $0-(-31)$

6. Avval javob musbat yoki manfiy bo'lishini ayting, so'ng hisoblang.
   a) $(-45)+(-17)$ b) $63+(-28)$ c) $(-52)+19$ d) $-36-(-50)$

7. Tugmalar bilan tasvirlab hisoblang. Har birida nechta nol juftlik qo'shish kerak?
   a) $1-4$ b) $(-2)-3$ c) $(-6)-(-4)$ d) $2-(-3)$

8. Ifodaning qiymatini toping.
   a) $6-(-9)+(-11)$ b) $(-4)-(-10)+(-8)$ c) $(-13)+(-5)-(-18)$
   d) $9-(-3)+(-14)$ e) $-20-(-16)+7$ f) $(-8)+(-6)+(-1)$

9. Hisoblang.
   a) $-9+2-8$ b) $-5-12+(-6)$ c) $7-(+15)+4$ d) $20-(-4)-(-9)$

10. $\square$ o'rniga qaysi son yoziladi?
    a) $5+\square=-4$ b) $-3+\square=9$ c) $-6-\square=-15$
    d) $2-\square=10$ e) $\square-(-8)=-2$ f) $-4+\square-(-3)=-11$

11. Do'kondagi muzqaymoq vitrinasida harorat −22 °C edi. Vitrina qopqog'i ochiq qolib ketdi va termometr −9 °C ni ko'rsatdi. Harorat necha gradusga ko'tarilgan?

12. Oy sirtida kunduzi harorat taxminan +120 °C gacha ko'tariladi, kechasi esa −170 °C gacha tushadi. Kunduzgi va tungi harorat farqi necha gradus?

13. Telefon balansida 3 500 so'm bor edi. Operator qarzga gaplashishga ruxsat beradi. Abdulla 6 000 so'mlik gaplashdi, keyin balansni 10 000 so'mga to'ldirdi.
    a) Gaplashgandan keyin balans qancha bo'ldi? b) To'ldirgandan keyin-chi?

14. Shaxta lifti yer sathidan 120 m pastda turgan edi. U 75 m ko'tarilib, keyin 40 m tushdi. Endi lift yer sathiga nisbatan qayerda?

15. Futbol jamoasining o'yinlari: 3:1, 0:2, 1:4, 2:2, 5:1 (birinchi son — urgan, ikkinchisi — o'tkazib yuborgan gollar).
    a) Har o'yinning gol farqini butun son bilan yozing (urgan − o'tkazgan).
    b) Beshta o'yindagi jami gol farqini toping.
    c) Oltinchi o'yindan keyin jami gol farqi −2 bo'ldi. Oltinchi o'yinning gol farqi qancha?

16. Toshkent GMT+5, Los-Anjeles (qishda) GMT−8 vaqt mintaqasida. Toshkentlik Sevara Los-Anjelesdagi xolasi bilan videoqo'ng'iroqda gaplashmoqchi. Xolasi 2-mart kuni o'z vaqti bilan soat 19:00 dan keyin bo'sh.
    a) Ikki shahar vaqti necha soat farq qiladi?
    b) Sevara Toshkent vaqti bilan qaysi kuni, soat nechadan keyin qo'ng'iroq qila oladi?

17. Ikki g'avvosdan biri dengiz sathidan 14 m, ikkinchisi 37 m chuqurlikda. a) Ular orasidagi tik masofani toping. b) Ikkinchi g'avvos 9 m yuqoriga ko'tarildi. Endi masofa qancha?

18. Rostmi yoki yolg'on? Yolg'on bo'lsa, qarshi misol keltiring.
    a) Musbat son − manfiy son = musbat son b) Manfiy son + manfiy son = musbat son
    c) Musbat son − musbat son = musbat son d) 0 − manfiy son = musbat son

19. −14 dan 15 gacha bo'lgan barcha butun sonlarning yig'indisini toping: $(-14) + (-13) + \cdots + 14 + 15$.

20. Jadvalda har bir qator, har bir ustun va ikkala diagonal bo'yicha sonlar yig'indisi −3 ga teng. Bo'sh kataklarni to'ldiring.

−4		0
	−1	
	−3	

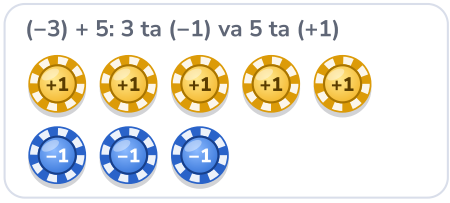
5-QISM

Tekshiramiz

1. Hisoblang: a) $(-16) + (-9)$ b) $21 + (-30)$ c) $(-8) + 27$ d) $0 - (-12)$

2. Hisoblang: a) $5 - 14$ b) $(-11) - 9$ c) $(-3) - (-17)$ d) $13 - (-6)$

3. Rasmda $(-3) + 5$ tugmalar bilan tasvirlangan. Nechta nol juftlik olib tashlanadi? Yig'indi nechaga teng?



4. Ifodaning qiymatini toping: a) $-7 + 15 - (-4)$ b) $10 - (-2) + (-20)$ c) $(-9) - (-9) + (-9)$

5. $\square$ ni toping: a) $-8 + \square = -3$ b) $\square - 6 = -10$ c) $4 - \square = 12$

6. Tog' cho'qqisida ertalab harorat −11 °C edi. Kunduzi havo 5 °C ga isidi, kechqurun 9 °C ga sovidi. Kechqurun harorat necha gradus?

7. O'lik dengiz qirg'og'i dengiz sathidan taxminan 430 m pastda, Toshkent esa taxminan 450 m balandda. Ularning balandliklari farqi qancha?

8. Lima (Peru) GMT−5 vaqt mintaqasida. Toshkentda (GMT+5) soat 07:00 bo'lganda Limada soat nechi bo'ladi?

9. «Ayirma har doim kamayuvchidan kichik» — rostmi? Javobni misol bilan asoslang.

10. **Xatoni toping.**
    a) Sardor: $-6-(-4)=-10$. b) Dilnoza: $-9+4=-13$. c) Kamola: $8-(-3)=5$.

- [ ] Men butun sonlarni algebraik tugmalar va nol juftliklar yordamida qo'sha va ayira olaman.
- [ ] Men yig'indining ishorasini oldindan ayta olaman.
- [ ] Men ayirishni qarama-qarshi sonni qo'shish bilan almashtira olaman: $a-(-b)=a+b$.
- [ ] Men harorat, balandlik, pul va vaqt mintaqalariga oid masalalarni yecha olaman.

Javoblar

4-qism · Mashqlar

1-masala

Butun sonlar: $\frac{12}{4} = 3$, −5, 0, −9,0 $= -9$, 100. Butun emas: 3,6; $-2\frac{1}{2}$; $\frac{7}{3}$.

2-masala

a) 3 ta juft ketadi, 2 ta -1 qoladi: -2. b) Hammasi nol juftlik: 0. c) 2 ta juft ketadi, 4 ta $+1$ qoladi: 4.

3-masala

a) −14. b) 23. c) 0. d) $-(-9) = 9$, qarama-qarshisi −9.

4-masala

a) 15. b) −15. c) −7. d) 9. e) 7. f) −19. g) 0. h) −14.

5-masala

a) −7. b) −15. c) $8 + 5 = 13$. d) $-14 + 20 = 6$. e) 0. f) −32. g) $-22 + 13 = -9$. h) 31.

6-masala

a) Manfiy: −62. b) Musbat: 35. c) Manfiy: −33. d) $-36 + 50$ — musbat: 14.

7-masala

a) 3 ta nol juftlik, 4 ta $+1$ olinadi: −3. b) 3 ta nol juftlik, 3 ta $+1$ olinadi: −5. c) Nol juftlik kerak emas, 4 ta -1 olinadi: −2. d) 3 ta nol juftlik, 3 ta -1 olinadi: 5.

8-masala

a) $6 + 9 - 11 = 4$. b) $-4 + 10 - 8 = -2$. c) $-13 - 5 + 18 = 0$. d) $9 + 3 - 14 = -2$. e) $-20 + 16 + 7 = 3$. f) −15.

9-masala

a) −15. b) −23. c) $7 - 15 + 4 = -4$. d) $20 + 4 + 9 = 33$.

10-masala

a) $-4-5=-9$. b) $9-(-3)=12$. c) $-6+15=9$. d) $2-10=-8$. e) $-2+(-8)=-10$. f) $-4+\square+3=-11$, keyin $-1+\square=-11$, $\square=-10$.

11-masala

$-9-(-22)=-9+22=13$: harorat 13 °C ga ko'tarilgan.

12-masala

$120-(-170)=120+170=290$ °C.

13-masala

a) $3500-6000=-2500$ so'm (qarz). b) $-2500+10\,000=7500$ so'm.

14-masala

$-120+75-40=-85$: yer sathidan 85 m pastda.

15-masala

a) $+2$; -2; -3; 0; $+4$. b) $2-2-3+0+4=1$. c) $1+\square=-2$, $\square=-2-1=-3$.

16-masala

a) $5-(-8)=13$ soat, Toshkent oldinda. b) $19{:}00+13$ soat: $19{:}00+5$ soat $=24{:}00$, yana 8 soat — keyingi kun 08:00. Sevara 3-mart soat 08:00 dan keyin qo'ng'iroq qila oladi.

17-masala

a) $-14-(-37)=23$ m. b) Ikkinchisi $-37+9=-28$ m da: $-14-(-28)=14$ m.

18-masala

a) Rost: musbat songa musbat son qo'shiladi. b) Yolg'on: $(-2)+(-3)=-5$. c) Yolg'on: $2-5=-3$. d) Rost: $0-(-a)=a$.

19-masala

Qarama-qarshi sonlarni juftlaymiz: $(-14)+14=0$, $(-13)+13=0$, ..., $(-1)+1=0$; 0 ham yig'indini o'zgartirmaydi. Faqat 15 qoladi: yig'indi 15.

20-masala

1-qator: $-3-(-4)-0=1$. Diagonallar: $-3-(-4)-(-1)=2$ (pastki o'ng), $-3-0-(-1)=-2$ (pastki chap). 1-ustun: $-3-(-4)-(-2)=3$, 3-ustun: $-3-0-2=-5$.

Javob: 1-qator −4, 1, 0; 2-qator 3, −1, −5; 3-qator −2, −3, 2.

5-qism · Tekshiramiz

1-masala

a) −25. b) −9. c) 19. d) 12.

2-masala

a) −9. b) −20. c) $-3+17=14$. d) $13+6=19$.

3-masala

3 ta nol juftlik olib tashlanadi, 2 ta $+1$ qoladi: $(-3)+5=2$.

4-masala

a) $-7+15+4=12$. b) $10+2-20=-8$. c) $-9+9-9=-9$.

5-masala

a) $-3-(-8)=5$. b) $-10+6=-4$. c) $4-12=-8$.

6-masala

$-11+5-9=-15$ °C.

7-masala

$450-(-430)=880$ m.

8-masala

Farq $5-(-5)=10$ soat. $07{:}00-10$ soat — kechagi kun soat 21:00.

9-masala

Yolg'on: $3 - (-2) = 5$, 5 esa 3 dan katta. Manfiy sonni ayirsak, natija kattalashadi.

10-masala

a) $-6 - (-4) = -6 + 4 = -2$. b) $-9 + 4 = -5$ (manfiylar ko'p, 9 – 4 = 5). c) $8 - (-3) = 8 + 3 = 11$.